Math Prize for Girls 2025 — P3

March 24, 2026

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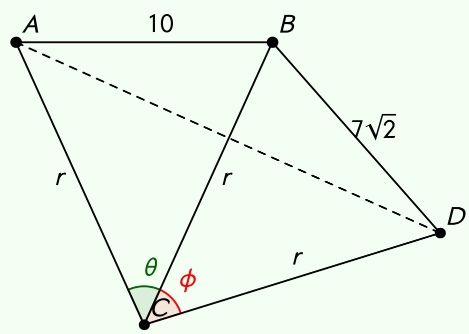
An octagon is inscribed in a circle. It has four consecutive sides of length 10 and four consecutive sides of length $7\sqrt{2}$. What is the radius of the circle?

Solution

Let O be the centre of the circle. Suppose that the sides of length 10 subtend an angle of θ at O, and that the sides of length $7\sqrt{2}$ subtend an angle of ϕ. Because there are 4 sides of length 10, $7\sqrt{2}$ we have

$$4\theta + 4\phi = 360 \iff \theta + \phi = 90$$

Consider,



Because of the two isoceles triangles we know that

$$\angle ABC + \angle CBD = \left(90 - \frac{\theta}{2}\right) + \left(90 - \frac{\phi}{2}\right) = 180 - 45 = 135.$$

So we can use the cosine rule to find out AD,

$$AD^2 = 10^2 + (7\sqrt{2})^2 - 2(10)(7\sqrt{2})\cos 135 = 338.$$

Now since $\theta + \phi = 90$, $\triangle ACD$ is right-angled and thus

$$2r^2 = x^2 = 338 \implies r = \boxed{13}.$$